

KIRENSK, Russian Federation

WMO: 302300

Lat: 57.773N

Lon: 108.059E

Elev: 259

StdP: 98.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.6	(c) -43.1	(d) -47.9	(e) 0.0	(f) -45.6	(g) -45.4	(h) 0.0	(i) -43.1	(j) 7.9	(k) -13.8	(l) 7.0	(m) -13.8	(n) 0.2	(o) 210	(p) 0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	29.9	19.0	28.0	18.5	26.0	17.7	20.7	26.6	19.7	25.4	18.7	24.2	2.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.0	23.1	17.6	13.0	22.3	16.6	12.2	21.2	61.1	26.7	57.8	25.5	54.3	24.3	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.3	5.4	-47.3	33.2	2.7	1.5	-49.3	34.2	-50.9	35.1	-52.4	36.0	-54.4	37.1		
			-47.6	23.5	2.9	1.6	-49.7	24.6	-51.4	25.5	-53.0	26.4	-55.1	27.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-26.4	-21.6	-11.7	-0.7	7.5	15.9	18.6	15.5	7.2	-1.9	-15.9	-25.1		
	DBStd	17.30	10.06	8.89	7.64	4.90	4.13	4.04	3.22	3.52	4.03	5.46	10.32	10.64		
	HDD10.0	5451	1127	884	671	321	101	4	0	4	104	370	778	1087		
	HDD18.3	7909	1386	1118	929	570	337	95	36	102	335	628	1028	1346		
	CDD10.0	663	0	0	0	1	23	179	268	173	19	0	0	0		
	CDD18.3	79	0	0	0	0	1	21	46	12	0	0	0	0		
	CDH23.3	1087	0	0	0	0	26	361	521	174	5	0	0	0		
	CDH26.7	306	0	0	0	0	4	104	156	42	0	0	0	0		
	WSAvg	2.0	1.5	1.6	2.1	2.6	2.6	2.1	1.8	1.7	2.0	2.3	2.0	1.5		
(15) Precipitation	PrecAvg	382	19	14	10	15	34	47	67	60	40	27	27	26		
	PrecMax	513	37	34	25	42	82	102	180	116	72	68	46	47		
	PrecMin	278	6	4	0	2	7	4	9	7	1	1	11	9		
	PrecStd	60	7	6	6	8	14	24	35	28	16	13	9	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.9	0.1	8.5	16.2	27.3	31.7	32.7	30.3	23.8	12.6	4.9	-1.8		
		MCWB	-4.0	-2.0	2.2	7.2	14.2	18.8	20.6	19.5	15.5	7.4	3.1	-3.0		
	2%	DB	-5.6	-2.3	4.5	13.1	23.2	29.6	30.3	27.5	19.3	9.5	2.5	-4.2		
		MCWB	-6.4	-4.0	0.2	5.6	11.9	18.2	20.0	18.7	12.8	5.7	0.7	-5.1		
	5%	DB	-8.4	-4.9	1.8	10.1	20.1	27.5	28.2	25.1	16.2	7.2	0.8	-6.4		
		MCWB	-9.1	-6.1	-1.4	4.0	10.2	17.6	19.1	17.8	11.4	4.4	-0.3	-7.1		
	10%	DB	-11.4	-8.1	-0.1	7.4	16.9	25.2	26.3	22.8	13.8	4.9	-1.3	-9.1		
		MCWB	-12.0	-9.1	-2.7	2.4	8.8	16.5	18.6	16.6	10.2	2.6	-2.4	-9.7		
	0.4%	WB	-3.6	-1.9	2.7	8.0	14.9	21.0	22.7	21.1	16.3	8.1	3.6	-2.6		
		MCDB	-2.9	-0.2	7.5	14.6	24.9	28.0	29.1	27.0	22.1	11.0	4.6	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-6.4	-3.9	0.6	6.0	13.0	19.5	21.2	19.7	14.0	6.4	1.0	-5.1		
		MCDB	-5.6	-2.5	3.8	12.3	21.1	26.6	27.3	25.4	18.3	8.9	2.1	-4.3		
	5%	WB	-9.1	-6.2	-1.0	4.4	11.1	18.5	20.3	18.5	12.2	4.6	-0.4	-7.1		
		MCDB	-8.4	-5.0	1.7	9.4	17.8	25.3	26.0	23.9	15.0	6.7	0.9	-6.5		
	10%	WB	-11.9	-9.0	-2.6	2.9	9.6	17.3	19.4	17.3	10.6	2.9	-2.4	-9.7		
		MCDB	-11.3	-8.1	-0.2	6.8	16.0	23.6	24.7	21.8	13.3	4.8	-1.3	-9.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	13.1	15.9	12.0	12.7	13.2	11.7	11.6	9.5	8.0	8.7	9.1		
		MCDBR	9.8	12.8	15.8	15.9	19.3	17.9	16.7	16.4	14.7	11.4	7.2	9.0		
		MCWBR	9.4	11.4	12.0	9.1	9.3	7.8	7.7	8.1	8.2	7.6	6.2	8.6		
		MCDBR	10.1	12.8	15.2	14.4	16.6	15.4	13.8	14.5	12.3	10.2	7.0	9.3		
	5% WB	MCWBR	9.8	11.6	11.6	8.5	8.5	7.3	7.1	7.7	7.3	7.3	6.3	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.270	0.303	0.324	0.356	0.396	0.434	0.490	0.421	0.343	0.319	0.278	0.255		
	taud		1.995	2.008	2.051	2.134	2.241	2.222	2.064	2.247	2.444	2.327	2.065	1.958		
	Ebn at Noon		541	701	807	847	836	805	743	776	802	703	529	429		
	Edh at Noon		59	93	118	129	125	130	149	116	81	68	54	45		
	RadAvg		0.67	1.62	3.12	4.53	5.04	5.72	5.22	4.22	2.61	1.30	0.70	0.43		
(45) All-Sky Solar Radiation	RadStd		0.06	0.12	0.19	0.28	0.35	0.51	0.53	0.35	0.23	0.17	0.09	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page